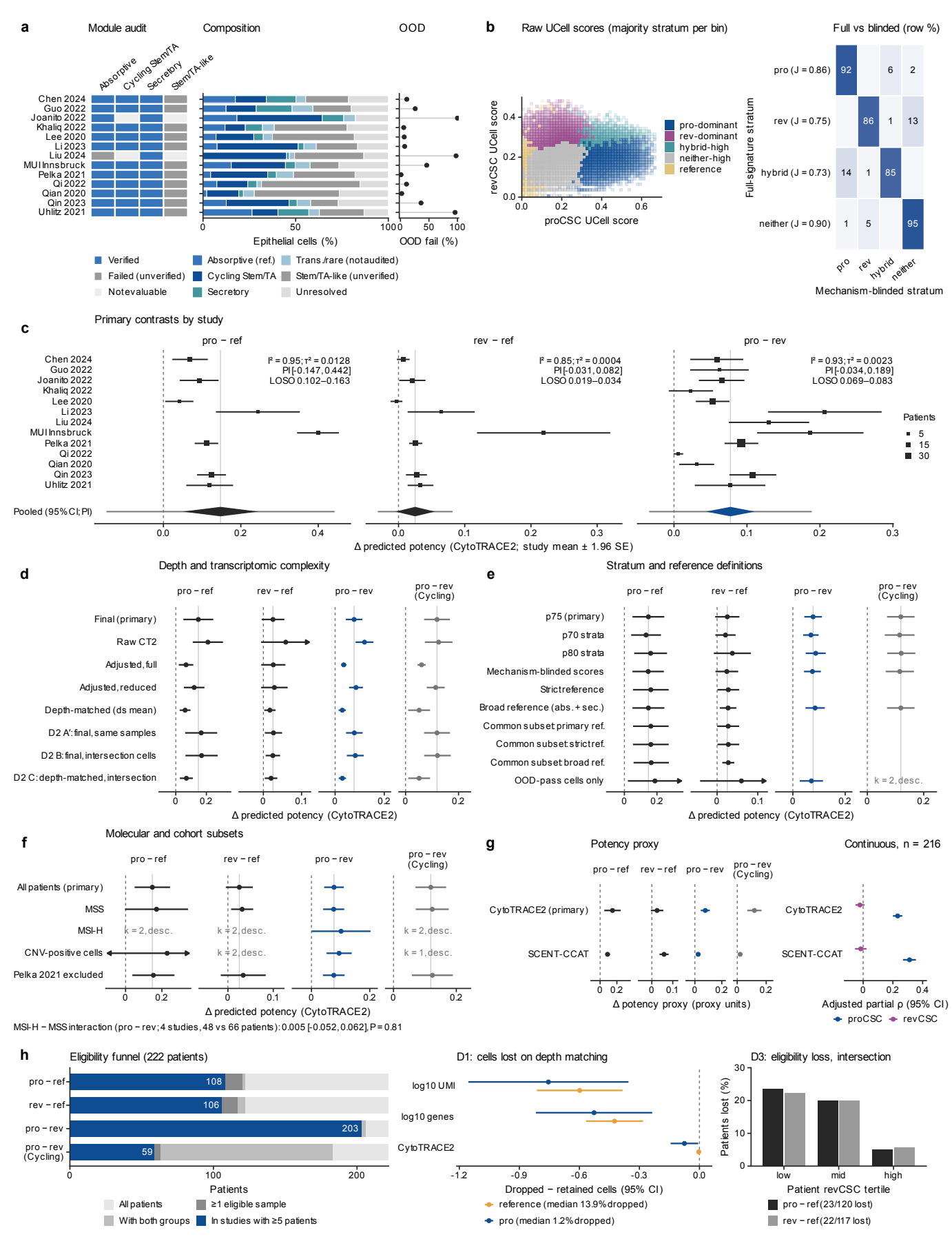

a
Module audit
Composition
OOD
Absorptive
Cycling Stem/TA
Secretory
Stem/TA-like
Chen 2024
Guo 2022
Joanito 2022
Khaliq 2022
Lee 2020
Li 2023
Liu 2024
MUI Innsbruck
Pelka 2021
Qi 2022
Qian 2020
Qin 2023
Uhlitz 2021
Epithelial cells (%)
OOD fail (%)
Verified
Failed (unverified)
Not evaluable
Absorptive (ref.)
Cycling Stem/TA
Secretory
Stem/TA-like
Unresolved
Trans./rare (not audited)
b
Raw UCell scores (majority stratum per bin)
revCSC UCell score
proCSC UCell score
pro-dominant
rev-dominant
hybrid-high
neither-high
reference
Full vs blinded (row %)
Full-signature stratum
pro (J = 0.86)
rev (J = 0.75)
hybrid (J = 0.73)
neither (J = 0.90)
92
6
2
86
1
13
14
1
85
1
5
95
pro
rev
hybrid
neither
Mechanism-blinded stratum
c
Primary contrasts by study
pro − ref
rev − ref
pro − rev
I² = 0.95; τ² = 0.0128
PI [-0.147, 0.442]
LOSO 0.102–0.163
I² = 0.85; τ² = 0.0004
PI [-0.031, 0.082]
LOSO 0.019–0.034
I² = 0.93; τ² = 0.0023
PI [-0.034, 0.189]
LOSO 0.069–0.083
Patients
5
15
30
Pooled (95% CI; PI)
Δ predicted potency (CytoTRACE2; study mean ± 1.96 SE)
d
Depth and transcriptomic complexity
pro − ref
rev − ref
pro − rev
pro − rev (Cycling)
Final (primary)
Raw CT2
Adjusted, full
Adjusted, reduced
Depth-matched (ds mean)
D2 A′: final, same samples
D2 B: final, intersection cells
D2 C: depth-matched, intersection
Δ predicted potency (CytoTRACE2)
e
Stratum and reference definitions
p75 (primary)
p70 strata
p80 strata
Mechanism-blinded scores
Strict reference
Broad reference (abs. + sec.)
Common subset: primary ref.
Common subset: strict ref.
Common subset: broad ref.
OOD-pass cells only
k = 2, desc.
Δ predicted potency (CytoTRACE2)
f
Molecular and cohort subsets
pro − ref
rev − ref
pro − rev
pro − rev (Cycling)
All patients (primary)
MSS
MSI-H
CNV-positive cells
Pelka 2021 excluded
k = 2, desc.
k = 1, desc.
Δ predicted potency (CytoTRACE2)
MSI-H − MSS interaction (pro − rev; 4 studies, 48 vs 66 patients): 0.005 [-0.052, 0.062], P = 0.81
g
Potency proxy
Continuous, n = 216
pro − ref
rev − ref
pro − rev
pro − rev (Cycling)
CytoTRACE2 (primary)
SCENT-CCAT
Δ potency proxy (proxy units)
CytoTRACE2
SCENT-CCAT
Adjusted partial ρ (95% CI)
proCSC
revCSC
h
Eligibility funnel (222 patients)
pro − ref
rev − ref
pro − rev
pro − rev (Cycling)
108
106
203
59
Patients
All patients
≥1 eligible sample
With both groups
In studies with ≥5 patients
D1: cells lost on depth matching
log10 UMI
log10 genes
CytoTRACE2
Dropped − retained cells (95% CI)
reference (median 13.9% dropped)
pro (median 1.2% dropped)
D3: eligibility loss, intersection
Patients lost (%)
low
mid
high
Patient revCSC tertile
pro − ref (23/120 lost)
rev − ref (22/117 lost)